
Continuity

Work these in class. Worked solutions are on the posted notes page — each problem below names the section heading to look for.

2.5 • Problem 1*see: What Continuity Says*

What are the three? Two of them are about the two sides of that equation existing at all.

2.5 • Problem 2*see: Three Ways to Fail*

For each picture, say which of the three requirements fails. Two of them fail the same one — which two, and why?

2.5 • Problem 3 (worked together)*see: Continuous Where?*

Is $\tan x$ continuous?

2.5 · Problem 4

see: Making It Continuous On Purpose

Work it. The three requirements tell you exactly what to write down.

2.5 · Problem 5

see: Why "Continuous" Is the Hypothesis

Sketch a function on $[0, 2]$ with $f(0) = -1$ and $f(2) = 1$ that *never* equals 0. What must your sketch have?

2.5 · Problem 6

see: Using the IVT: Showing a Root Exists

You are not being asked to solve it. What *are* you being asked for, and what would you have to check?